ArchiTexture: Recovering Texture Partitions from Frozen SAM via Proposal-Space Commitment

Anonymous Author(s)
Affiliation
Address
email

Abstract

Frozen SAM-style segmenters often fail on texture partitions, but failure does not by itself show that texture evidence is absent. We study texture segmentation as a recoverability problem: how much task value can be extracted from frozen features and frozen proposals before retraining the backbone or proposal generator. Our main contribution is proposal-space recovery, instantiated by ARCHITEXTURE, which keeps the proposal bank fixed and learns texture partition commitment through proposal compatibility scoring, conservative core selection, and an RWTD-specific dense rescue layer. We retain a smaller feature-space recovery branch only as auxiliary diagnostic evidence for latent texture structure in coarse frozen features. Under matched evaluation, ARCHITEXTURE improves RWTD common-253 coherence from 0.6163 to 0.7013 ARI against a public TextureSAM rerun while keeping mIoU comparable, and improves STLD common-182 from 0.5140/0.7526 to 0.7195/0.7791. Oracle analysis shows that the frozen bank still contains substantial unrecovered value: on RWTD the bank upper bound reaches 0.5183 mIoU / 0.8580 ARI. These results support a narrower claim: in frozen SAM-style texture segmentation, the main missing ingredient is often partition commitment above fragmented proposals rather than new texture features inside the backbone. Code: `https://github.com/scientific-computing-user/ArchiTexture`.

1 Introduction

Texture partitioning is not just semantic segmentation with unusual labels. The target regions are often defined by repeated local structure, appearance statistics, and transition behavior rather than by named object categories. RWTD and STLD make that distinction explicit, and they also expose the practical bottleneck: texture boundaries can be weak, disconnected, or globally ambiguous even when local evidence is strong Khan et al. [2015, 2017], Khan and Sundaramoorthi [2018], Mikeš and Haindl [2022]. SAM-style foundation segmenters add a new failure mode on top of that difficulty. They often emit many plausible masks on the same image, yet still fail to commit to the texture partition that matters for the task Kirillov et al. [2023], Ravi et al. [2024].

That failure is easy to misread. If a frozen SAM system misses the final texture partition, the default response is to assume that the backbone itself must be specialized for texture. TextureSAM is the strongest current example of that strategy on the RWTD/STLD benchmark family Cohen et al. [2025, 2026]. Our starting point is different. Fragmented masks suggest that the relevant evidence may already be present inside the frozen system and that the missing step is not always feature invention. It may instead be *recoverability*: whether the final texture partition can be recovered from frozen evidence.

Feature-space recovery	Proposal-space recovery
Frozen evidence: multiscale SAM `backbone_fpn` features.	**Frozen evidence:** SAM-style mask proposals.
Question: Is texture partition structure already latent in the representation?	**Question:** If the bank already contains the right fragments, can a learned layer recover a coherent partition without retraining the proposal generator?
Readout: pooled coarsest clustering, flip-averaging, positionality diagnostics.	**Readout:** compatibility reasoning, conservative core selection, and selective repair above the frozen bank.
Role in paper: diagnostic probe only.	**Role in paper:** main operational contribution instantiated by ARCHITEXTURE.

Figure 1: Two recoverability loci in frozen SAM-style systems. The feature route asks whether texture structure is already latent in frozen features and is used only as a diagnostic probe. The proposal route is the main operational question of the paper: how much task value can be recovered by learning commitment above a frozen proposal bank. Both routes keep the underlying segmenter frozen.

This paper asks where recoverable texture structure already lives inside a frozen SAM-style system. There are two natural loci. One is feature space: coarse frozen features may already contain texture partition structure. The other is proposal space: the frozen mask bank may already contain the right fragments, but the system may fail to assemble them into a coherent partition. The two loci play different roles here. Proposal-space recovery is the operational core of the paper. Feature-space recovery remains a diagnostic probe that explains why recoverability is plausible in the first place.

This framing separates two questions that are often conflated. One question is whether frozen foundation segmenters contain texture evidence at all. The other is whether that evidence has already been organized into the correct final partition. Our main empirical claim concerns the second question. ARCHITEXTURE keeps the proposal generator frozen and learns the commitment layer above it: which fragments are compatible, which merged component should be trusted as a conservative core, and on RWTD only, when a denser rescue candidate is worth the risk. Under matched evaluation, this proposal-space route improves RWTD coherence from 0.6163 to 0.7013 ARI on the common-253 comparison against TextureSAM while maintaining comparable mIoU, and it improves STLD from 0.5140/0.7526 to 0.7195/0.7791 on the common-182 direct evaluator.

The paper is therefore not a claim that backbone specialization is unnecessary. It is a narrower claim that frozen SAM systems often contain more texture evidence than their final outputs reveal. On RWTD, for example, the frozen bank upper bound reaches 0.5183 mIoU / 0.8580 ARI on the common-253 subset, well above the deployed system. That gap is exactly what makes proposal-space recoverability scientifically meaningful.

Contributions. The paper makes three focused contributions.

- We formulate texture segmentation in frozen SAM-style systems as a recoverability problem and make proposal-space commitment above a frozen proposal bank the paper's primary operational contribution.
- We show, on matched RWTD and STLD evaluations, that learned commitment above a frozen bank recovers substantial task value and that generic bank-only heuristics are not enough.
- We add oracle decompositions showing that the frozen bank is richer than the deployed decision layer, while keeping the feature-space branch only as a constrained diagnostic probe rather than as a competing main comparison.

2 Related Work

Texture segmentation and benchmarks. Classical texture recognition and segmentation work established the core datasets and problem structure that still matter here, including DTD, MINC, and the RWTD/STLD benchmark lineage Cimpoi et al. [2014], Bell et al. [2015], Cimpoi et al. [2016], Khan et al. [2015, 2017], Khan and Sundaramoorthi [2018], Mikeš and Haindl [2022]. Much of that literature focuses either on handcrafted/local descriptors or on fully supervised dense predictors specialized to a fixed texture regime Andrearczyk and Whelan [2017], Liu et al. [2019]. Our question is different: given a frozen foundation segmenter, how much texture partition structure is already recoverable before any backbone retraining?

SAM-family adaptation. Segment Anything made high-recall proposal banks widely available, and SAM 2 extended that foundation-segmentation interface to stronger image and video settings Kirillov et al. [2023], Ravi et al. [2024]. Most SAM adaptation work improves the foundation model itself through prompt tuning, adversarial tuning, or higher-quality decoders Ke et al. [2023], Zhang et al. [2023], Li et al. [2024]. TextureSAM is the most relevant prior route for this paper because it specifically injects texture behavior into the SAM backbone and establishes a strong baseline on the RWTD/STLD family Cohen et al. [2025, 2026]. We do not argue against that line. We ask whether recoverable texture behavior also lives above a frozen proposal engine.

Representation probing and proposal reasoning. A small clustering-based probe can answer a different question from a task-trained decoder: not "can a large head solve the benchmark," but "what structure is already latent in the frozen representation?" That perspective motivates our feature-space branch. Our proposal-space branch instead connects texture segmentation to a more general class of reasoning problems over candidate masks: compatibility scoring, conservative selection, and selective completion from a frozen proposal bank. In this paper those two views are unified rather than treated as unrelated methods. The feature route asks where texture structure lives in the representation; the proposal route asks how much task value can be recovered from the proposals that the frozen system already emits.

3 Problem Formulation

Let a frozen SAM-style segmenter expose two kinds of evidence for an image x: a multiscale feature representation $\phi(x)$ and a proposal bank $\mathcal{P}(x) = \{m_i\}_{i=1}^{N_x}$. The downstream target is a binary texture partition $Y \in \{0,1\}^{H\times W}$. Depending on the benchmark, the foreground side may be canonical (STLD) or label-symmetric (RWTD and the appendix routes), but the output is always one binary partition.

We define *recoverability* as the performance of a readout that must keep the underlying segmenter frozen. The readout may inspect $\phi(x)$, $\mathcal{P}(x)$, or both, but it may not retrain the backbone or alter the proposal generator. This yields two loci:

$$\hat{Y}_{\mathrm{feat}} = R_{\mathrm{feat}}(\phi(x)), \tag{1}$$

$$\hat{Y}_{\mathrm{prop}} = R_{\mathrm{prop}}(x, \mathcal{P}(x)). \tag{2}$$

The first readout asks whether texture structure is already latent in frozen features. The second asks whether fragmented proposal evidence can be assembled into a coherent task partition.

For the proposal route, the frozen bank is first encoded proposal-wise as $d_i = E(x, m_i)$. A learned pair model scores potentially adjacent proposals,

$$s(i,j) = f_{\mathrm{pair}}(\psi(d_i, d_j, m_i, m_j)), \tag{3}$$

and connected components induced by scores above a threshold τ define candidate merged regions $\mathcal{C}(x)$. A learned component scorer then ranks those candidates,

$$q(C) = f_{\mathrm{comp}}(g(C; x, \mathcal{P}(x))), \qquad C^\star(x) = \arg\max_{C\in\mathcal{C}(x)} q(C). \tag{4}$$

RWTD adds an optional denser rescue bank $\mathcal{P}^+(x)$ and a learned switch that decides whether some rescue candidate $R_k \in \mathcal{P}^+(x)$ should replace the conservative core $C^\star(x)$.

Representation-level recoverability. The feature route is intentionally lightweight as an engineered system. It clusters coarse frozen `backbone_fpn` features, optionally after pooling or flip-averaging, and then scores the resulting binary partition. Its role is diagnostic: if such a simple probe already succeeds at all, then the backbone is not texture-blind. Because the available RWTD feature artifacts use a 227-image public split rather than the official 256-image proposal-route evaluator, this branch is kept secondary throughout.

Route	Frozen evidence	Learned readout	Evaluator	Role in paper
Feature probe	multiscale SAM `backbone_fpn` features	none; clustering-only probe	available route-specific splits	diagnostic only; appendix evidence for latent structure, not a headline comparison
RWTD	frozen prompt bank plus dense rescue candidates	compatibility, conservative core ranking, rescue switch	official invariant mIoU / ARI on full-256 and common-253	main natural benchmark and main operational claim
STLD	frozen prompt bank	Stage-A compatibility and component ranking	direct foreground mIoU / ARI on all-200 and common-182	main synthetic companion benchmark
ControlNet bridge	frozen prompt bank	Stage-A compatibility and component ranking	invariant mIoU / ARI on all-1742	supporting appendix route only; no causal bridge-training claim
CAID	frozen prompt bank	Stage-A compatibility and component ranking	invariant mIoU / ARI on all-3104	appendix breadth check only

Table 1: Frozen-versus-learned summary of the routes used in the paper. Proposal-space recovery on RWTD and STLD carries the main claim; feature-space evidence is diagnostic only, and the bridge and CAID routes are supporting appendix material.

Proposal-level recoverability. The proposal route treats the frozen bank as incomplete evidence. The readout must decide which fragments are mutually compatible, which merged component should be trusted as the conservative core, and, on RWTD only, when a denser completion is worth the risk. This is the main operational setting of the paper. If a learned commitment layer can recover a strong final partition from frozen proposals, then texture failure in the frozen system does not imply texture absence in the bank.

Fair comparison. Recoverability claims are only meaningful when evaluator and subset conventions are fixed within each block. The main paper therefore uses matched comparisons only: RWTD on the official full-256 evaluator and on the common-253 subset shared with the public TextureSAM rerun, and STLD on the all-200 and common-182 direct-foreground views. The feature probe is not mixed into those tables.

4 Feature-Space Recovery

The feature-space branch asks the narrower question first: if SAM is kept frozen, does texture partition structure already exist in the representation at all? The repository contains a probe based on clustering pooled coarsest `backbone_fpn` features, with coarse-only and flip-averaged variants and documented positional-debiasing controls. That probe matters scientifically because it is the simplest possible readout: no decoder is trained, and the backbone remains untouched.

We treat that route conservatively. The available RWTD feature probe is evaluated on a 227-image public split, not on the official 256-image evaluator used by the proposal route, and we did not recover a matched rerun script that would make the comparison fully protocol-aligned. We therefore do *not* use feature-space numbers as a main comparison in the paper. Appendix B reports the repository-backed parity summary, a curated qualitative gallery, and scale-weighting diagnostics only to support a modest claim: coarse frozen features appear to contain nontrivial texture structure, and the probe is sensitive to positional debiasing such as flip-averaging.

That asymmetry is deliberate. The paper does not need the feature route to win a leaderboard. It only needs the feature route to motivate why recoverability above a frozen system is plausible. The main empirical burden is carried instead by matched proposal-space evaluation.

5 Proposal-Space Recovery

The proposal-space branch treats frozen automatic masks as fragmented evidence rather than as final predictions. This is the setting where ARCHITEXTURE operates. Given a frozen proposal bank, the system must decide which fragments belong together, which unions are contradictory, and whether a conservative core should be expanded. The key architectural point is simple: the proposal generator stays frozen throughout. What is learned is only the commitment layer above that bank.

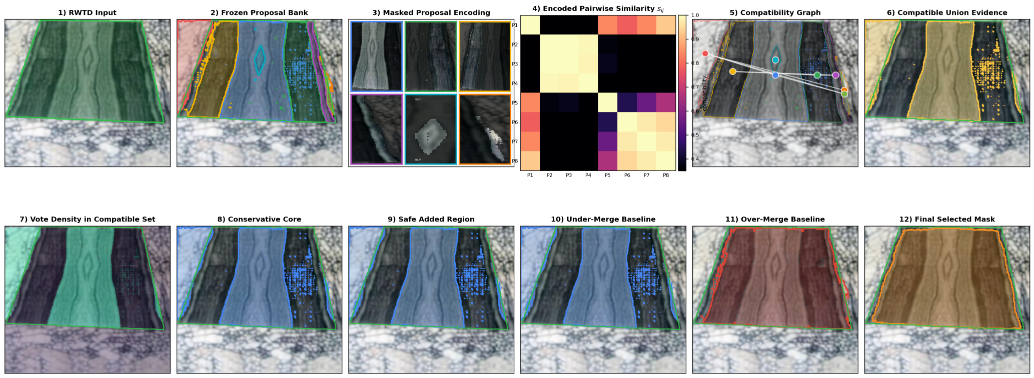

Figure 2: Proposal-space recovery in ARCHITEXTURE. Each frozen proposal m_i is encoded as a hybrid texture descriptor d_i, a learned pair model merges compatible proposals into candidate components, a learned component scorer selects a conservative core $C^\star$, and on RWTD only a learned dense-rescue switch can replace $C^\star$ with a denser candidate. The gain therefore comes from commitment above a fixed bank, not from a stronger proposal generator.

The implementation makes the learned objects explicit. Each proposal m_i is encoded by a hybrid descriptor

$$d_i = [h_i; e_i], \tag{5}$$

where h_i is a 60-D handcrafted summary of a 15-channel training-free texture map and e_i is a PTD descriptor built from a masked crop plus ring context. The PTD encoder produces a 512-D region embedding, a second 512-D ring-context contrast term, and concatenates them into a 1024-D descriptor. The handcrafted map concatenates LAB color, Sobel $x/y/\|\nabla\|$, a Laplacian response, and an 8-filter Gabor bank; h_i stores the mean, standard deviation, lower quartile, and upper quartile of those channels inside the proposal mask. Stage-A therefore uses a 1084-D descriptor per proposal.

Proposal compatibility is learned with a RANDOMFORESTCLASSIFIER over a 10-D pair feature vector. That vector includes descriptor cosine similarity, proposal area ratios, pair IoU, near-contact statistics, weak-boundary evidence, texture heterogeneity of the union, centroid distance, and union area ratio. Positive pair labels come from PTD-synthetic compositions: two proposals are compatible only when they cover the same synthetic texture region with sufficient purity. Thresholding these pair scores produces a compatibility graph, and its connected components define the candidate merged regions.

Candidate components are then ranked by a RANDOMFORESTREGRESSOR over a second 10-D feature vector. Those component features include region area ratio, inside-versus-outside texture contrast, within-region variance, fragmentation penalty, boundary strength, compactness, descriptor variance within the component, descriptor separation from the complement, and proposal-count statistics. The regressor is trained to predict candidate IoU against the synthetic ground-truth partition, and the highest-scoring candidate becomes the conservative core $C^\star$.

RWTD adds one more learned decision layer above that core. A denser auxiliary bank proposes rescue candidates R_k around under-covered scenes. For each $(C^\star, R_k)$ pair, the RWTD rescue path builds a core-versus-candidate feature vector that includes predicted score gaps, relative area and overlap, proposal-union coverage and precision, texture contrast, variance, and fragmentation deltas. A balanced logistic classifier predicts whether switching to R_k is a safe positive move under `oracle_safe_winner` labels, and a HISTGRADIENTBOOSTINGREGRESSOR predicts the expected gain. The deployed `candidate_cls` rule keeps the core unless some rescue candidate is classified positive, in which case it selects the positive candidate with highest predicted gain. STLD and the appendix routes stop at Stage-A and never invoke this rescue layer. In notation,

$$p_{\text{safe}}(k) = f_{\text{safe}}(u(C^\star, R_k)), \tag{6}$$
$$g_{\text{gain}}(k) = f_{\text{gain}}(u(C^\star, R_k)), \tag{7}$$

where $u(C^\star, R_k)$ is the rescue feature vector. The deployed prediction is

$$\hat{Y}_{\text{prop}} = \begin{cases} R_{k^\star}, & \exists k : p_{\text{safe}}(k) > \tau_{\text{safe}},\ k^\star = \arg\max_{k:p_{\text{safe}}(k)>\tau_{\text{safe}}} g_{\text{gain}}(k), \\ C^\star, & \text{otherwise.} \end{cases} \tag{8}$$

Training regime and frozen components. The proposal generator is frozen. ARCHITEXTURE does not retrain SAM, alter the prompt source, or optimize the bank with RWTD labels. What is learned instead is the decision stack above the bank. The RWTD Stage-A bundle used here is trained on 480 PTD-only synthetic samples, and the RWTD rescue bundle is trained on 256 PTD-only `repair_mix` samples that inject explicit under-coverage cores and hard wrong-side decoys. The STLD Stage-A bundle is trained on 1600 PTD-only MPEG7-shape synthetic layouts. This separation is why the method speaks directly to recoverability: it measures how much task value remains once proposal generation is treated as fixed evidence rather than as the object of adaptation.

Why this model family? The decision layer is deliberately small. The goal of the paper is to isolate recoverability above a frozen bank, not to bury it inside a new end-to-end network. The available supervision is limited and synthetic, the proposal descriptors mix heterogeneous geometric and texture cues, and the RWTD rescue path must remain modular enough to turn on only where it is needed. Tree-based models fit that regime well: they combine mixed tabular signals robustly, remain easy to audit at the level of pair scores and candidate rankings, and let the proposal bank stay fixed throughout. Appendix A summarizes the deployed inference algorithm, descriptor ablations, and the resulting practicality profile.

6 Datasets and Protocol

The paper's main scope is intentionally narrow. RWTD is the main natural benchmark and remains evaluation-only. STLD is the controlled synthetic companion with a canonical foreground side. Two additional routes remain in the appendix: a ControlNet-stitched bridge benchmark with exact binary supervision and a CAID breadth check. Neither route drives the main claim.

RWTD. RWTD is the paper's anchor benchmark because it concentrates the failure modes that motivate recoverability: weak transitions, disconnected same-texture regions, and repeated-pattern distractors. Main results use the official 256-image evaluator and, for direct comparison to the public TextureSAM rerun, the common-253 subset of images where both systems emit predictions.

STLD. STLD is the main synthetic companion. It is useful here because the shaped foreground side is canonical and exact supervision is available. That makes it a cleaner test of whether proposal-space commitment above a frozen bank can recover the intended partition when evaluator ambiguity is minimal.

Supporting routes. The appendix keeps two scope-limited routes. The ControlNet bridge benchmark is a controlled binary partition route generated by a ControlNet-conditioned stitched-regeneration pipeline; it preserves exact labels but uses more naturalistic transitions than classical hard mosaics. We use it as supporting evidence rather than a headline pillar because this paper does not establish a causal bridge-training effect. CAID is included as a domain-shift breadth check because its shoreline focus is less central to the paper's texture claim.

Metrics and subset conventions. RWTD uses official mIoU and ARI under complementary binary partitions. STLD uses direct foreground mIoU and ARI because the shaped foreground side is canonical. The appendix bridge and CAID routes use partition-invariant mIoU and ARI. Every headline table keeps evaluator and subset conventions fixed within each comparison block; unmatched legacy numbers are not mixed into the main results.

7 Results

We organize the results around the central recoverability questions. First, does a learned commitment layer recover task value above a frozen proposal bank under matched evaluation? Second, how

Benchmark	Evaluator / subset	Method	mIoU	ARI
RWTD	official invariant, full-256	SAM2.1-small original rerun Ravi et al. [2024]	0.1615	0.2183
RWTD	official invariant, full-256	ArchiTexture final	0.4611	0.6966
RWTD	official invariant, common-253	TextureSAM public checkpoint rerun TextureSAM Authors [2026]	0.4684	0.6163
RWTD	official invariant, common-253	ArchiTexture final	0.4645	0.7013
STLD	direct foreground, all-200	SAM2.1-small original rerun Ravi et al. [2024]	0.3686	0.5269
STLD	direct foreground, all-200	ArchiTexture final	0.6705	0.7249
STLD	direct foreground, common-182	TextureSAM public checkpoint rerun TextureSAM Authors [2026]	0.5140	0.7526
STLD	direct foreground, common-182	ArchiTexture final	0.7195	0.7791

Table 2: Main matched comparisons. Each block fixes the evaluator and subset convention before comparing methods. Supporting bridge and CAID routes are reported separately in the appendix so that the headline table remains protocol-matched.

much of the answer is already present in that bank? Third, which part of the decision stack is actually responsible for the gain? The feature probe remains secondary and is not used as a headline comparison in this section.

Proposal-space recovery under matched evaluation. The headline comparison is simple. On RWTD, the raw SAM2.1-small baseline is far below either texture-specialized adaptation or decision-layer recovery above a frozen bank. Against the matched TextureSAM rerun on the common-253 subset, ArchiTexture reaches comparable overlap quality while improving partition coherence from 0.6163 to 0.7013 ARI. Because the official RWTD no-aggregation metric averages matched GT instances rather than simple per-image scores, we summarize uncertainty on the same overlap-instance view used by the paired significance analysis: on the 426 overlapped GT instances, the gain over TextureSAM is $+0.0995$ ARI with a 95% paired bootstrap interval of $[0.0721,\ 0.1270]$, while mIoU changes by $+0.0405$ $[0.0031,\ 0.0764]$. The full-256 deployment supports the same reading: the method remains far above the raw SAM baseline at 0.4611 mIoU / 0.6966 ARI.

STLD shows the same pattern under a cleaner direct-foreground evaluator. On the common-182 subset shared with the public TextureSAM rerun, ArchiTexture improves from 0.5140 / 0.7526 to 0.7195 / 0.7791. A paired bootstrap on the per-image outputs gives a mean mIoU gain of +0.2055 with a 95% interval of [0.1558, 0.2518] on the common-182 subset and +0.2028 [0.1555, 0.2486] on the all-200 view. The ARI gain is smaller but remains positive on average in both views. Together, RWTD and STLD establish the operational claim of the paper: substantial texture task value can be recovered without retraining the proposal generator. The practical cost is measurable rather than hidden: Appendix A and Table 7 show that RWTD Stage-A runs at 0.025 s/image on the measured GPU profile, while the full dense-rescue deployment costs 0.378 s/image and peaks at 637 MiB RSS.

How much evidence is already in the frozen bank? The bank-oracle analysis is the core evidence for the recoverability framing. Figure 3 and Table 3 show that on RWTD the frozen bank is much richer than the deployed decision layer. A single frozen proposal already reaches 0.5142 mIoU / 0.8146 ARI on the common-253 subset, and the bank upper bound reaches 0.5183 / 0.8580. The learned single-proposal selector reaches only 0.4512 / 0.5601 on the same subset: it improves over the generic medoid single baseline, but it still trails the full commitment stack by 0.1412 ARI. This means the main gap is not proposal absence alone, and it is not just top-1 proposal ranking either. It is the quality of commitment above a bank that often already contains usable fragments.

The same qualitative pattern appears on STLD, but with a useful twist. On the common-182 subset, a learned single-proposal selector already reaches 0.7196 mIoU / 0.7731 ARI, nearly matching the full system on overlap and trailing by only 0.0060 ARI. The single frozen-proposal oracle reaches 0.8105 / 0.8444, while the bank upper bound reaches 0.8149 / 0.8574. By contrast, the compatible-component oracle and top-k union oracle are both poor. The cross-dataset contrast is informative: STLD often contains a coherent singleton answer, whereas RWTD more often requires fragment commitment and selective repair. Table 13 makes that regime difference explicit. RWTD is primarily a fragmented-evidence problem, while STLD is often a singleton-selection problem. That contrast strengthens the recoverability framing rather than weakening it: commitment above the

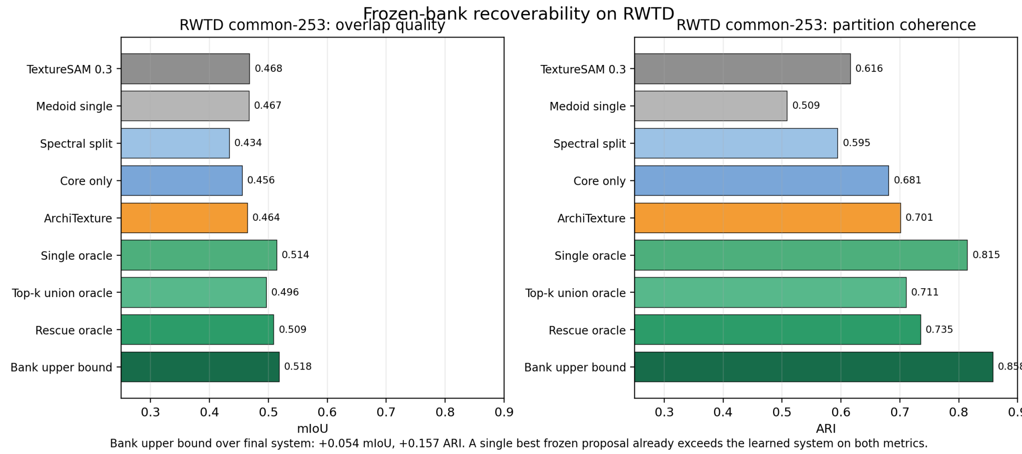

Figure 3: RWTD common-253 recoverability decomposition. Generic frozen-bank heuristics are noticeably weaker than the learned commitment layer, but the bank oracles remain substantially higher than the final deployed system. The frozen bank therefore already contains more texture evidence than the final decision layer recovers.

RWTD common-253 method	mIoU	ARI
Medoid single	0.4673	0.5087
Learned single selector	0.4512	0.5601
Spectral split	0.4339	0.5948
Agglomerative split	0.4354	0.5898
Greedy merge	0.4442	0.5197
Core only	0.4558	0.6812
ARCHITEXTURE final	0.4645	0.7013
Single frozen-proposal oracle	0.5142	0.8146
Top-k union oracle	0.4964	0.7107
Rescue candidate oracle	0.5093	0.7352
Bank upper bound	0.5183	0.8580

Table 3: RWTD recoverability decomposition on the matched common-253 subset. The learned top-1 selector is stronger than the medoid single baseline, but it still trails the full commitment stack by a wide ARI margin. The bank oracles remain far higher still. The omitted compatible-component oracle performs poorly (0.4748 mIoU / 0.2497 ARI), which is evidence that compatibility merging alone is not sufficient.

bank matters most exactly where the frozen evidence arrives as pieces rather than as one already coherent proposal.

How much does the descriptor matter? Appendix A and Table 6 isolate the descriptor-sensitive Stage-A layer while keeping the frozen bank, evaluator, and model family fixed. The descriptor choice matters most on RWTD: the hybrid descriptor improves Stage-A coherence to 0.6812 ARI on common-253, versus 0.6556 for handcrafted-only and 0.6505 for PTD-only. STLD is less sensitive and even slightly favors handcrafted-only on ARI, although hybrid remains the strongest mIoU setting there. The point is not that descriptors are irrelevant. The point is that the proposal-space claim does not collapse to one lucky cue stack: richer hybrid cues help most in the harder fragmented RWTD regime, while the same commitment formulation remains meaningful across descriptor families.

Which decisions matter? The route-level ablations isolate why ARCHITEXTURE is not just a larger union or a generic graph rule. On RWTD, the medoid single baseline nearly matches the final system on mIoU but collapses ARI, and even the learned top-1 selector remains far below the full system on coherence. The conservative core already recovers much of that ARI gap, and the final system adds the remaining improvement through selective rescue. On STLD, proposal union, handcrafted consolidation, and the PTD heuristic are all far below the learned system on the common-182 direct evaluator, while the learned top-1 selector is already very strong. The supporting

Route	Variant	mIoU	ARI
RWTD common-253	Medoid single (generic bank baseline)	0.4673	0.5087
RWTD common-253	Learned single selector	0.4512	0.5601
RWTD common-253	Conservative core only	0.4558	0.6812
RWTD common-253	ARCHITEXTURE final	0.4645	0.7013
STLD common-182	Proposal union	0.2543	0.1731
STLD common-182	Handcrafted consolidation	0.3680	0.3987
STLD common-182	PTD heuristic	0.3757	0.4085
STLD common-182	Learned single selector	0.7196	0.7731
STLD common-182	ARCHITEXTURE final	0.7195	0.7791
ControlNet bridge all-1742	Proposal union (supporting route)	0.6749	0.1188
ControlNet bridge all-1742	ARCHITEXTURE Stage-A (supporting route)	0.6803	0.6039

Table 4: Route-specific ablations using each route's native evaluator. On RWTD, learned top-1 selection helps but remains well below the structured commitment stack, while the conservative core already captures most of the coherence gain and the final rescue layer adds the remaining improvement. On STLD, by contrast, learned top-1 selection is already very strong, which highlights the difference between canonical synthetic partitions and the more fragmented natural RWTD scenes.

bridge route shows the same coherence effect under its own invariant metric: proposal union has reasonable overlap but very poor ARI, while the Stage-A learned system preserves coherence.

Residual failure modes. The proposal-space story is strong but not complete. The full RWTD audit still records wrong-partition commitment on 4 of 256 images, rescue activations on 21 images, and top-k recoverability on 141 of 256 images. In other words, the remaining gap is not that the bank never contains the answer. The remaining gap is ambiguity handling: how to preserve multiple plausible partitions long enough to avoid brittle winner-take-all commitment. That limitation is exactly what the oracle gap exposes. Appendix C expands this point with selector/core/final/oracle case studies, and Appendix D shows the complementary STLD regime where a coherent singleton often already exists.

8 Discussion and Limitations

The paper makes one claim dominant: proposal-space commitment above a frozen bank is a strong and measurable source of recoverable texture performance. The feature-space branch remains useful, but only as background diagnosis. That asymmetry is a strength, not a weakness. It lets us say something more precise than either "SAM already solves texture" or "SAM is texture-blind." Frozen SAM-style systems often contain relevant evidence, but they do not reliably organize it into the right final partition.

The oracle gap makes that statement concrete. On both RWTD and STLD, the bank upper bound is well above the deployed system, and a single frozen proposal often outperforms the final learned output. At the same time, naive unions and naive compatibility merges do not solve the problem. The missing ingredient is not just access to more masks. It is a decision rule that can preserve coherence without collapsing to the wrong partition in ambiguous scenes.

The limitations are equally clear. First, the feature-space branch remains protocol-limited: the available RWTD feature evidence comes from a public 227-image split rather than the official proposal-route evaluator, so the paper does not use it as a main comparison. Second, the appendix bridge route is not a central claim because the manuscript does not establish a causal bridge-training benefit. Third, CAID remains a breadth check rather than a core texture benchmark. Finally, the learned RWTD rescue layer is still imperfect in low-margin scenes. The next step is therefore not necessarily a different frozen backbone. It is better ambiguity handling above the bank: calibration, abstention, or multi-hypothesis outputs that avoid premature commitment.

9 Reproducibility Note

Every headline number is traceable to a rerun or a named artifact in `EXPERIMENT_LEDGER.md` and `UPDATED_SOURCES_USED.md`. RWTD remains evaluation-only throughout the paper: no main claim depends on RWTD-label training or hyperparameter search. The manuscript workspace is self-contained, so paper compilation does not depend on older manuscript trees or external figure paths.

References

Vincent Andrearczyk and Paul F. Whelan. Texture segmentation with fully convolutional networks. *arXiv preprint arXiv:1703.05230*, 2017. doi: 10.48550/arXiv.1703.05230. URL `https://arxiv.org/abs/1703.05230`.

Sean Bell, Paul Upchurch, Noah Snavely, and Kavita Bala. Material recognition in the wild with the materials in context database. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3479–3487, 2015. doi: 10.1109/CVPR.2015.7298970. URL `https://openaccess.thecvf.com/content_cvpr_2015/papers/Bell_Material_Recognition_in_2015_CVPR_paper.pdf`.

Mircea Cimpoi, Subhransu Maji, Iasonas Kokkinos, Sammy Mohamed, and Andrea Vedaldi. Describing textures in the wild. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3606–3613, 2014. doi: 10.1109/CVPR.2014.461. URL `https://openaccess.thecvf.com/content_cvpr_2014/html/Cimpoi_Describing_Textures_in_2014_CVPR_paper.html`.

Mircea Cimpoi, Subhransu Maji, Iasonas Kokkinos, and Andrea Vedaldi. Deep filter banks for texture recognition, description, and segmentation. *International Journal of Computer Vision*, 118(1):65–94, 2016. doi: 10.1007/s11263-015-0872-3. URL `https://link.springer.com/article/10.1007/s11263-015-0872-3`.

Inbal Cohen, Boaz Meivar, Peihan Tu, Shai Avidan, and Gal Oren. Texturesam: Towards a texture aware foundation model for segmentation, 2025. URL `https://arxiv.org/abs/2505.16540`. arXiv preprint.

Inbal Cohen, Boaz Meivar, Peihan Tu, Shai Avidan, and Gal Oren. Decoupling shape and texture in SAM-2 via controlled texture replacement. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026. URL `https://openaccess.thecvf.com/content/WACV2026/papers/Cohen_Decoupling_Shape_and_Texture_in_SAM-2_via_Controlled_Texture_Replacement_WACV_2026_paper.pdf`.

Lei Ke, Mingqiao Ye, Martin Danelljan, Yifan Liu, Yu-Wing Tai, Chi-Keung Tang, and Fisher Yu. Segment anything in high quality. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL `https://proceedings.neurips.cc/paper_files/paper/2023/file/5f828e38160f31935cfe9f67503ad17c-Paper-Conference.pdf`.

Naeemullah Khan and Ganesh Sundaramoorthi. Learned shape-tailored descriptors for segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. doi: 10.1109/CVPR.2018.00076. URL `https://openaccess.thecvf.com/content_cvpr_2018/papers/Khan_Learned_Shape-Tailored_Descriptors_CVPR_2018_paper.pdf`.

Naeemullah Khan, Marei Algarni, Anthony Yezzi, and Ganesh Sundaramoorthi. Shape-tailored local descriptors and their application to segmentation and tracking. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3890–3899, 2015. doi: 10.1109/CVPR.2015.7299014. URL `https://openaccess.thecvf.com/content_cvpr_2015/html/Khan_Shape-Tailored_Local_Descriptors_2015_CVPR_paper.html`.

Naeemullah Khan, Byung-Woo Hong, Anthony Yezzi, and Ganesh Sundaramoorthi. Coarse-to-fine segmentation with shape-tailored continuum scale spaces. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4240–4249, 2017. doi: 10.1109/CVPR.2017.188. URL `https://openaccess.thecvf.com/content_cvpr_2017/html/Khan_Coarse-To-Fine_Segmentation_With_CVPR_2017_paper.html`.

Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloé Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3992–4003, 2023. doi: 10.1109/ICCV51070.2023.00371. URL `https://openaccess.thecvf.com/content/ICCV2023/papers/Kirillov_Segment_Anything_ICCV_2023_paper.pdf`.

Bo Li, Haoke Xiao, and Lv Tang. ASAM: Boosting segment anything model with adversarial tuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3699–3710, 2024. doi: 10.48550/arXiv.2405.00256. URL `https://openaccess.thecvf.com/content/CVPR2024/papers/Li_ASAM_Boosting_Segment_Anything_Model_with_Adversarial_Tuning_CVPR_2024_paper.pdf`.

Li Liu, Jie Chen, Paul Fieguth, Guoying Zhao, Rama Chellappa, and Matti Pietikainen. From BoW to CNN: Two decades of texture representation for texture classification. *International Journal of Computer Vision*, 127(1):74–109, 2019. doi: 10.1007/s11263-018-1125-z. URL `https://link.springer.com/article/10.1007/s11263-018-1125-z`.

Stanislav Mikeš and Michal Haindl. Texture segmentation benchmark. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(9):5647–5663, 2022. doi: 10.1109/TPAMI.2021.3075916. URL `https://doi.org/10.1109/TPAMI.2021.3075916`.

Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Hend Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Trevor Darrell, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer. SAM 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024. doi: 10.48550/arXiv.2408.00714. URL `https://arxiv.org/abs/2408.00714`.

TextureSAM Authors. TextureSAM public checkpoint artifact, 2026. Publicly available checkpoint and prediction artifact used for reproduced baseline evaluation.

Renrui Zhang, Zhengkai Jiang, Ziyu Guo, Shilin Yan, Junting Pan, Dong Hao, Yu Qiao, Peng Gao, and Hongsheng Li. Personalize segment anything model with one shot. *arXiv preprint arXiv:2305.03048*, 2023. doi: 10.48550/arXiv.2305.03048. URL `https://arxiv.org/abs/2305.03048`.

Algorithm 1: ARCHITEXTURE proposal-space inference for image x

1. Load the frozen prompt-bank proposals $\mathcal{P}(x)$ and compute proposal descriptors d_i for each $m_i \in \mathcal{P}(x)$.
2. Score adjacent proposal pairs with $s(i, j)$ and threshold the compatibility graph.
3. Form candidate merged regions from the graph connected components.
4. Score each candidate with $q(C)$ and choose the conservative core $C^\star = \arg\max_C q(C)$.
5. If the route is RWTD, build dense rescue candidates R_k from the auxiliary bank $\mathcal{P}^+(x)$.
6. Score each $(C^\star, R_k)$ with the safe-switch classifier $p_{\text{safe}}(k)$ and gain regressor $g_{\text{gain}}(k)$.
7. Output the positive rescue candidate with highest predicted gain, if any; otherwise output $C^\star$.

Table 5: Compact pseudocode for the deployed decision layer. STLD and the appendix routes stop after Step 4 because they do not use the RWTD rescue layer.

Benchmark / subset	Output	Handcrafted-only		PTD-only		Hybrid	
		mIoU	ARI	mIoU	ARI	mIoU	ARI
RWTD common-253	Stage-A official	0.4469	0.6556	0.4486	0.6505	0.4558	0.6812
RWTD full-256	Stage-A official	0.4454	0.6533	0.4472	0.6484	0.4543	0.6786
STLD common-182	Stage-A direct	0.7045	0.7808	0.6698	0.7740	0.7195	0.7791
STLD all-200	Stage-A direct	0.6570	0.7265	0.6253	0.7203	0.6705	0.7249

Table 6: Descriptor ablation for the descriptor-sensitive Stage-A commitment layer, with the proposal bank, evaluator, and model family held fixed. RWTD is reported at Stage-A because the dense rescue layer operates on candidate masks and is unchanged across descriptor variants. Hybrid descriptors are clearly strongest on RWTD, while STLD is less sensitive to the descriptor choice and even slightly favors handcrafted-only on ARI.

A Implementation Details and Practicality

Why include this appendix section? The paper's main claim is scientific rather than systems-oriented, but two reviewer questions still matter: whether the gain depends mainly on a particular descriptor stack, and whether the decision layer is practical enough to deploy above a frozen bank. The tables in this section answer those questions directly. The descriptor ablation keeps the decision stack fixed while varying only the proposal descriptor in the descriptor-sensitive Stage-A layer; on RWTD, hybrid descriptors help clearly more than either single family, while STLD is noticeably less sensitive and sometimes slightly favors handcrafted cues on ARI. The practicality table then summarizes the actual bank sizes, rescue scale, and measured runtime profile of the deployed RWTD and STLD routes.

B Feature-Space Diagnostic Evidence

The feature-space branch remains auxiliary. Its repository-backed artifacts come from the coarse-feature clustering bundles, the parity summaries, and the scale-probe study in the `research-site` tree. The key caveat is short and fixed throughout this appendix: the RWTD feature probe uses a 227-image public split rather than the official 256-image proposal-route evaluator, so this route is reported only as diagnostic evidence for where texture structure can already live inside frozen features.

Interpretation. The appendix diagnostics point in one direction. Coarse frozen features already recover much of the binary partition on the repository-backed probe, learned all-scale fusion still prefers the coarsest level, and the finest native grid is the least reliable setting. That is enough for the paper's narrower use of this route: frozen SAM features can contain texture-aware structure, but the feature probe remains a diagnostic instrument rather than a matched main-paper system.

Route	Prompt props	Merged comps	Dense props	Rescue cand.	s / img	Peak RSS	Reading
RWTD Stage-A	3.84	2.64	–	–	0.025	430 MiB	prompt-bank commitment only
RWTD full	3.84	2.64	38.05	11.72	0.378	637 MiB	dense rescue evaluated on the full set; 21/256 images switch
STLD Stage-A	1.78	2.32	–	–	–	–	no rescue path; no separate runtime profile recorded

Table 7: Practicality profile of the deployed decision layer. RWTD timings come from measured full-dataset GPU profiles of the same frozen-bank pipeline. STLD shares the same Stage-A architecture and bank statistics, but no separate runtime profile was recorded in the repository.

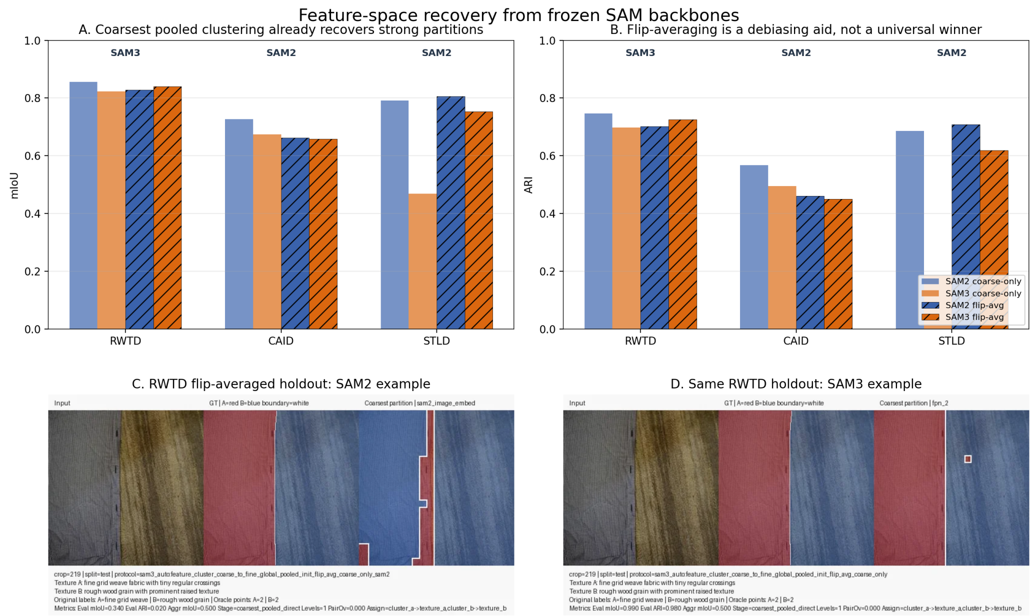

Figure 4: Archived feature-probe summary from the repository parity bundles. Top: coarse-only and flip-averaged summaries across RWTD, STLD, and CAID. Bottom: representative RWTD holdouts showing that coarse clustering can recover plausible binary partitions while remaining sensitive to positional bias.

C RWTD Oracle and Failure Details

RWTD is where the recoverability claim is most demanding. The proposal bank is strong, but the deployed decision stack still leaves a large oracle gap. Figure 7 shows four recurring patterns. First, top-1 proposal scoring can miss cases where conservative component commitment succeeds. Second, rescue can repair an under-covered core. Third, a strong singleton can already exist in the bank even when the deployed stack abstains from it. Fourth, the remaining wrong-partition failures are low-margin commitment mistakes rather than proposal-bank misses.

How to read the oracle gap. The bank upper bound is not a deployable method: it assumes access to the ground-truth partition when choosing among proposals, merged components, and rescue candidates. Its role is diagnostic. On RWTD, the gap between the deployed system and that bound shows that the frozen bank already contains more usable texture evidence than the current decision layer recovers. The learned single-selector baseline narrows only part of that gap, which is why RWTD still needs structured commitment above fragmented proposals.

D STLD Supporting Oracle Analysis

STLD plays a different supporting role. The oracle analysis still shows extra headroom in the bank, but the learned top-1 selector is already close to the full system on the common-182 subset. In

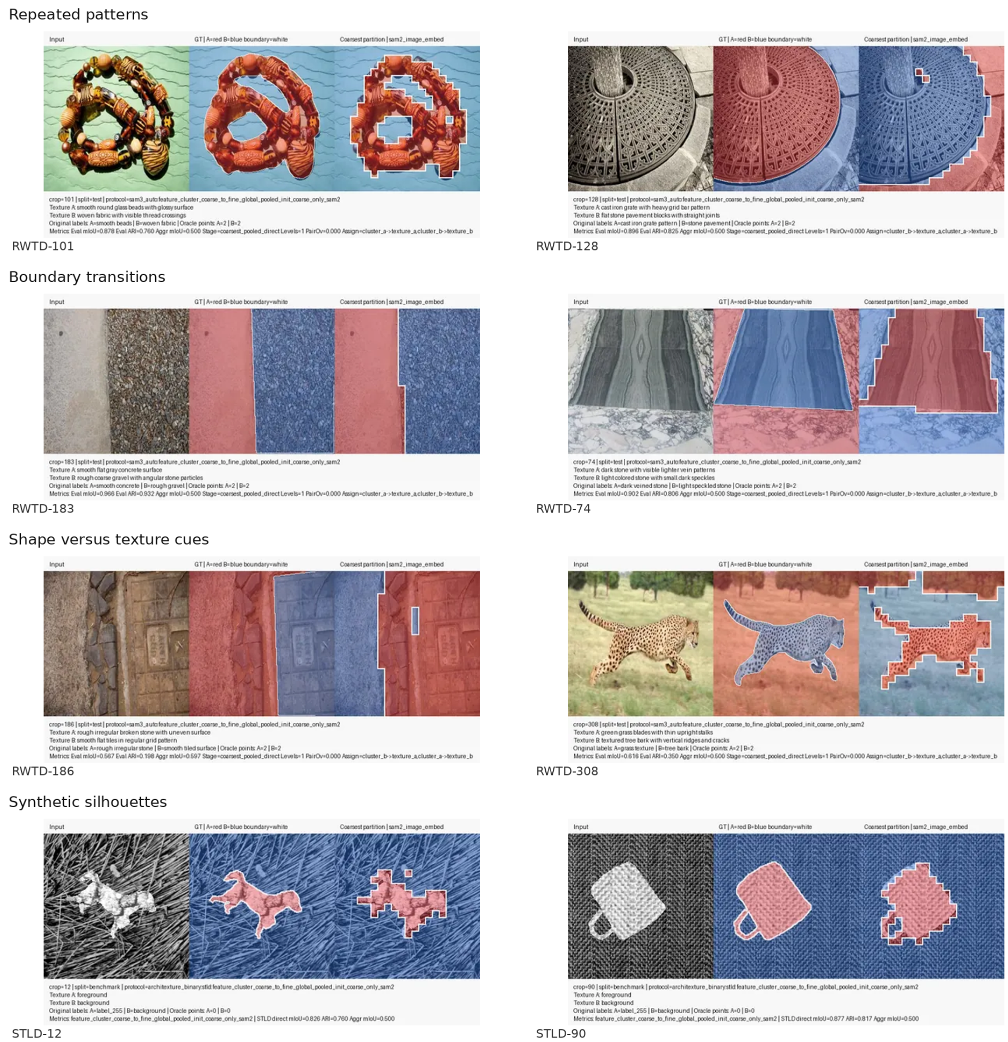

Figure 5: Curated qualitative gallery for coarse frozen-feature clustering. The gallery spans repeated-pattern scenes, gradual texture transitions, shape-versus-texture cases, and synthetic silhouettes. The annulus example in RWTD-101 is representative of the strongest qualitative evidence: even a frozen coarse probe can recover a topologically nontrivial partition without retraining the backbone.

practice, many STLD images contain a coherent singleton answer, so the main remaining gain is modest coherence cleanup rather than large-scale fragment commitment.

E Supporting Routes

The remaining routes stay secondary by design. The ControlNet-stitched bridge benchmark preserves exact supervision while introducing more naturalistic transitions than hard mosaics, but this paper does not claim a causal bridge-training benefit. CAID is kept only as a breadth check under domain shift.

E.1 ControlNet Bridge Examples

E.2 CAID Breadth Check

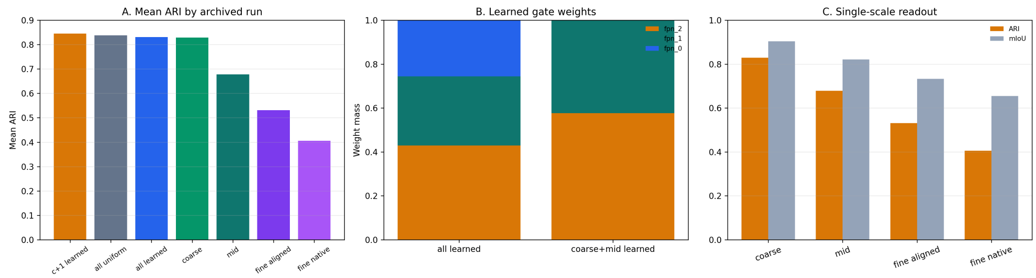

Figure 6: Archived scale and gating diagnostics from the feature probe. Left: mean ARI by run. Middle: learned all-scale and coarse-plus-mid gate weights still place most mass on the coarsest level. Right: single-scale readouts show that coarse features dominate, while native fine-grid clustering is the weakest control. These runs are diagnostic only and are not evaluated under the official RWTD proposal-route protocol.

Dataset	Eval view	SAM2 coarse-only	SAM3 coarse-only	SAM2 flip-avg	SAM3 flip-avg
RWTD	partition-invariant	0.8561 / 0.7472	0.8235 / 0.6984	0.8278 / 0.7013	0.8397 / 0.7258
CAID	partition-invariant	0.7263 / 0.5673	0.6743 / 0.4956	0.6620 / 0.4605	0.6588 / 0.4507
STLD	direct foreground	0.7908 / 0.6864	0.4682 / 0.1859	0.8064 / 0.7089	0.7534 / 0.6194

Table 8: Archived frozen-feature parity summary from the site bundle. Each cell is reported as mIoU / ARI. This table is diagnostic only and is not used as a matched comparison to the proposal-space evaluator.

Figure 7: RWTD case studies across selector, core, final, and oracle views. These examples make the oracle gap concrete: the bank often contains a strong answer, but the deployed stack still needs the right commitment decision to reach it. The rescue layer helps on some under-coverage cases, but low-margin wrong-partition failures remain brittle.

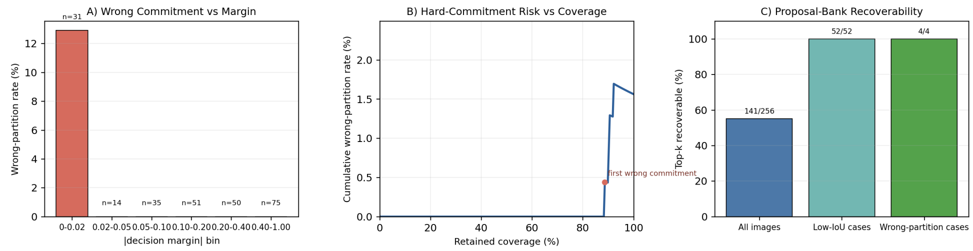

Figure 8: RWTD ambiguity-to-failure diagnostics on the full-256 audit. Wrong-partition cases concentrate in the lowest-margin regime, and top-k recoverability remains high exactly where the final decision layer is most brittle.

Setting	Resc.	Off-succ.	Hurt-any	Wrong	Unsafe	Mean ΔIoU	Top-k recov.
Baseline decision rule	21	61.9%	38.1%	4	6	+0.2247	141/256

Table 9: RWTD full-256 failure-audit summary for the deployed decision rule. The remaining weakness is ambiguity-aware commitment rather than proposal-bank absence.

Dense rescue proposal source (frozen)	Full-256 mIoU / ARI	Common-253 mIoU / ARI
Released dense bank	0.4611 / 0.6966	0.4645 / 0.7013
SAM2.1-large source swap	0.5003 / 0.7048	0.5005 / 0.7106

Table 10: Frozen proposal-source transfer stress test. Consolidation weights and thresholds remain fixed; only the rescue proposal bank changes. This is evidence for modularity of the decision layer, not a standalone benchmark claim.

Comparison	Paired units	Δ mIoU (95% interval)	Δ ARI (95% interval)
RWTD common-253, ArchiTexture − TextureSAM	426 overlapped GT instances	$+0.0405$ $[0.0031, 0.0764]$	$+0.0995$ $[0.0721, 0.1270]$
STLD all-200, ArchiTexture − TextureSAM	200 images	$+0.2028$ $[0.1555, 0.2486]$	$+0.0400$ $[0.0049, 0.0754]$
STLD common-182, ArchiTexture − TextureSAM	182 images	$+0.2055$ $[0.1558, 0.2518]$	$+0.0265$ $[-0.0097, 0.0605]$

Table 11: Paired interval support for the matched comparisons. RWTD uses the official overlap-instance view because the published no-aggregation metric averages GT instances overlapped by each method rather than simple per-image means. STLD intervals are paired per-image bootstraps under the direct-foreground evaluator.

Item	Value
External training data	PTD only for the RWTD route; Brodatz-native synthetic supervision for STLD
Encoder training data	80,000 train / 20,000 val images per synthetic source
RWTD benchmark usage	Evaluation only; no RWTD-label training or hyperparameter search
Proposal source	Frozen SAM-style automatic masks
RWTD/STLD deployed model	Swin-Base masked-region encoder with learned merge/core/repair modules
ControlNet / CAID deployed model	PTD-trained `small_pre_ring_hi` Stage-A bundle
Primary evaluator	Official upstream no-aggregation evaluator on RWTD; route-matched evaluators elsewhere
Comparator family	Raw SAM2.1-small plus reproduced/public TextureSAM checkpoints

Table 12: Compact reproducibility summary for the manuscript scope.

Dataset / subset	Selector mIoU	Selector ARI	Final mIoU	Final ARI	Reading
RWTD common-253	0.4512	0.5601	0.4645	0.7013	Large ARI gap: fragmented natural scenes still need structured commitment above the frozen bank.
STLD common-182	0.7196	0.7731	0.7195	0.7791	Near-parity: a coherent singleton answer often already exists, so commitment adds only a small coherence gain.

Table 13: Cross-dataset contrast between learned top-1 selection and full proposal-space commitment. RWTD and STLD use different evaluators, so the point is not absolute scale but the different role played by singleton selection on the two routes.

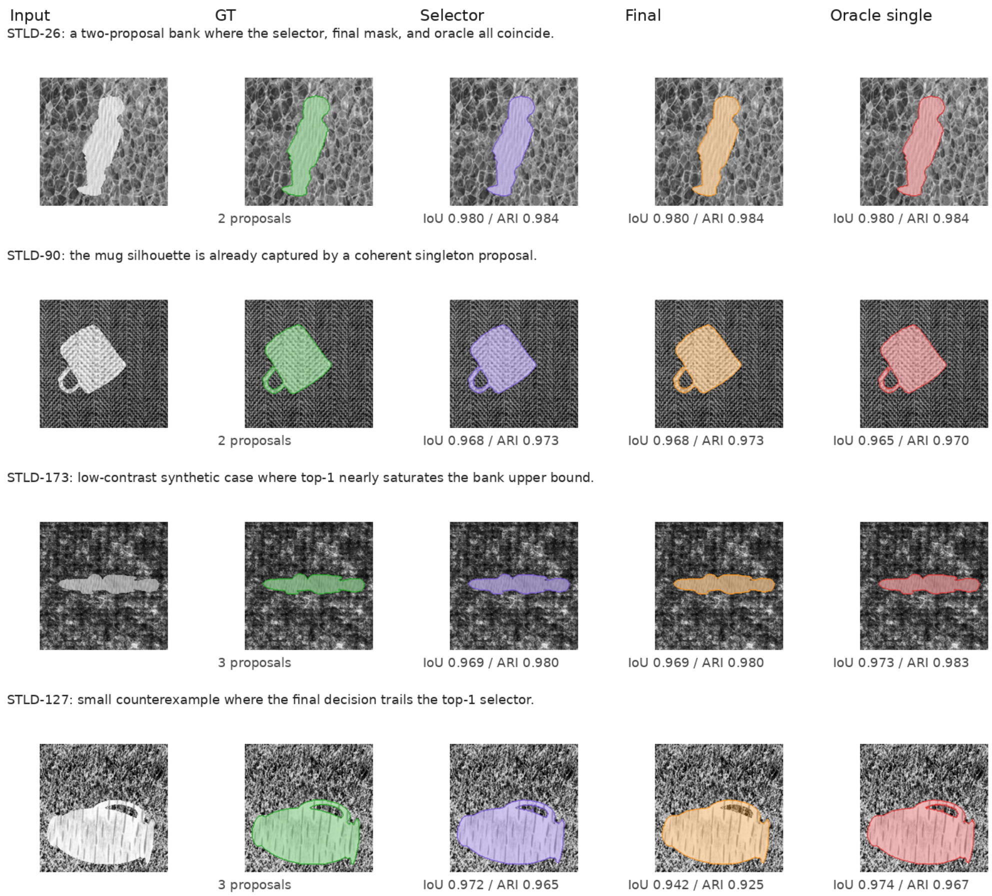

Figure 9: Representative STLD cases. Most examples already have a coherent singleton proposal, so the learned selector, final deployment, and single-proposal oracle are nearly indistinguishable. The last row shows a useful counterexample where the selector slightly outruns the final decision.

STLD method	All-200 mIoU	All-200 ARI	Common-182 mIoU	Common-182 ARI
ArchiTexture final	0.6705	0.7249	0.7195	0.7791
Learned single selector	0.6548	0.7036	0.7196	0.7731
Single frozen-proposal oracle	0.7375	0.7684	0.8105	0.8444
Compatible-component oracle	0.3950	0.3598	0.4341	0.3953
Top-k union oracle	0.1257	0.0289	0.1381	0.0317
Bank upper bound	0.7574	0.7962	0.8149	0.8574

Table 14: Supporting STLD oracle analysis under the direct-foreground evaluator. The learned top-1 selector already recovers much of the overlap on STLD, but the full system remains slightly stronger on coherence and still trails the bank oracles.

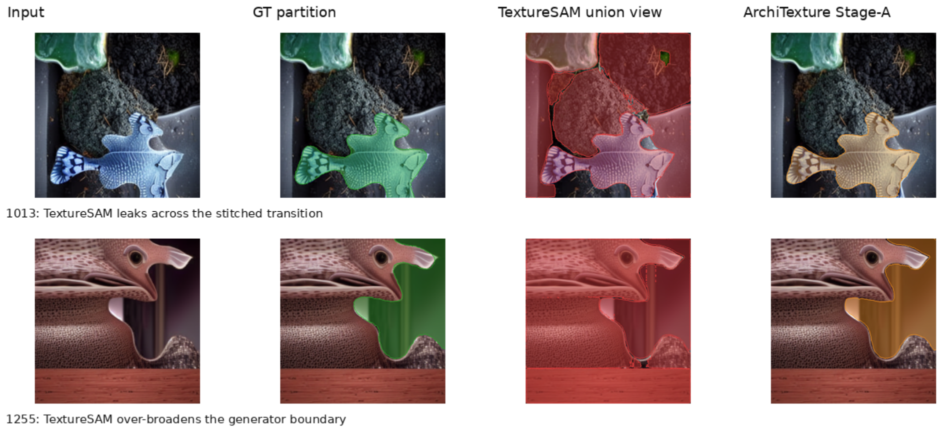

Figure 10: ControlNet-stitched bridge examples. The proposal-union view often leaks across the stitched transition or over-broadens the generator boundary, which explains why overlap can look reasonable while ARI remains fragile. The Stage-A route is more conservative, but we keep this benchmark in the appendix because the paper does not establish a causal bridge-training advantage.

Method	Subset / coverage	Direct mIoU / ARI	Invariant mIoU / ARI
TextureSAM public checkpoint rerun	all-1742, 1739/1742	0.4212 / 0.6446	0.6425 / 0.5510
ARCHITEXTURE Stage-A (ours)	all-1742, 1742/1742	0.4993 / 0.6039	0.6803 / 0.6039
TextureSAM public checkpoint rerun	common-1739	0.4219 / 0.6457	0.6436 / 0.5519
ARCHITEXTURE Stage-A (ours)	common-1739	0.4993 / 0.6039	0.6803 / 0.6039

Table 15: Supporting ControlNet-stitched bridge results. The main paper does not use this route as a central claim because the repository supports clean evaluation but not a causal bridge-training study.

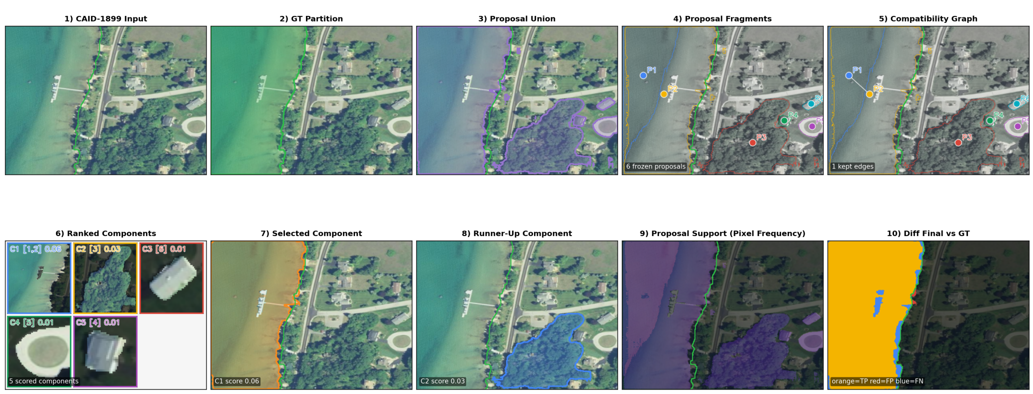

Figure 11: CAID breadth-check example. The same frozen-bank consolidation logic remains useful under shoreline structure, but this route is not central to the paper's texture claim and is reported only as an appendix-side generalization check.

Method	Subset / coverage	Invariant mIoU / ARI
TextureSAM public checkpoint rerun	all-3104, 3063/3104	0.6691 / 0.5080
ARCHITEXTURE Stage-A (ours)	all-3104, 3104/3104	0.6677 / 0.5745
TextureSAM public checkpoint rerun	common-3063	0.6780 / 0.5148
ARCHITEXTURE Stage-A (ours)	common-3063	0.6677 / 0.5745

Table 16: Supporting CAID breadth check under the partition-invariant evaluator. CAID is kept only as an appendix route because its shoreline focus is less central to the paper's texture claim.